

Cytochrome c Oxidase on the Crossroads of Transcriptional Regulation and Bioenergetics.

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Mitochondria are the organelles of eukaryotic cells responsible for the ATP production by means of the electron transfer chain (ETC). Its work is under strict genetic control providing the correct assembly of the enzyme complexes and the interface to adapt the energetic demands of the cell to the environment. These mechanisms are particularly developed in the cells with high energy consumption, like neurons and myocytes. This review summarizes several aspects of the involvement of the ETC complexes in the transcriptional control mechanisms of the neurons and other cells. Their influence on the differentiation of neurons is also discussed.

Balancing ER-Mitochondrial Ca²⁺ Fluxes in Health and Disease.

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Organelles cooperate with each other to control cellular homeostasis and cell functions by forming close connections through membrane contact sites. Important contacts are present between the endoplasmic reticulum (ER), the main intracellular Ca²⁺-storage organelle, and the mitochondria, the organelle responsible not only for the majority of cellular ATP production but also for switching on cell death processes. Several Ca²⁺-transport systems focalize at these contact sites, thereby enabling the efficient transmission of Ca²⁺ signals from the ER toward mitochondria. This provides tight control of mitochondrial functions at the microdomain level. Here, we discuss how ER-mitochondrial Ca²⁺ transfers support cell function and how their dysregulation underlies, drives, or contributes to pathogenesis and pathophysiology, with a major focus on cancer and neurodegeneration but also with attention to other diseases such as diabetes and rare genetic diseases.

The fate of damaged mitochondrial DNA in the cell.

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Mitochondrion is a double membrane organelle that is responsible for cellular respiration and production of most of the ATP in eukaryotic cells. Mitochondrial DNA (mtDNA) is the genetic material carried by mitochondria, which encodes some essential subunits of respiratory complexes independent of nuclear DNA. Normally, mtDNA binds to certain proteins to form a nucleoid that is stable in mitochondria. Nevertheless, a variety of physiological or pathological stresses can cause mtDNA damage, and the accumulation of damaged mtDNA in mitochondria leads to mitochondrial dysfunction, which triggers the occurrence of mitochondrial diseases in vivo. In response to mtDNA damage, cell initiates multiple pathways including mtDNA repair, degradation, clearance and release, to recover mtDNA, and maintain mitochondrial quality and cell homeostasis. In this review, we provide our current understanding of the fate of damaged mtDNA, focus on the pathways and mechanisms of removing damaged mtDNA in the cell.

The role of mitochondrial uncoupling proteins in lifespan.

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The increased longevity in modern societies raised the attention to biological interventions that could promote a healthy aging. Mitochondria are main organelles involved in the production of adenosine triphosphate (ATP), the energetic substrate for cellular biochemical reactions. The production of ATP occurs through the oxidative phosphorylation of intermediate substrates derived from the breakdown of lipids, sugars, and proteins. This process is coupled to production of oxygen reactive species (ROS) that in excess will have a deleterious role in cellular function. The damage promoted by ROS has been emphasized as one of the main processes involved in senescence. In the last decades, the discovery of specialized proteins in the mitochondrial inner membrane that promote the uncoupling of proton flux (named uncoupling proteins-UCPs) from the ATP synthase shed light on possible mechanisms implicated in the buffering of ROS and consequently in the process of aging. UCPs are responsible for a physiological uncoupling that leads to decrease in ROS production inside the mitochondria. Thus, induction of uncoupling through UCPs could decrease the cellular damage that occurs during aging due to excess of ROS. This review will focus on the evidence supporting these mechanisms.

Intracellular compartmentation of CTP synthase in *Drosophila*.

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Compartmentation is essential for the localization of biological processes within a eukaryotic cell. ATP synthase localizes to organelles such as mitochondria and chloroplasts. By contrast, little is known about the subcellular distribution of CTP synthase, the critical enzyme in the production of CTP, a high-energy molecule similar to ATP. Here I describe the identification of a novel intracellular structure containing CTP synthase, termed the cytoophidium, in *Drosophila* cells. I find that cytoophidia are present in all major cell types in the ovary and exist in a wide range of tissues such as brain, gut, trachea, testis, accessory gland, salivary gland and lymph gland. In addition, I find CTP synthase-containing cytoophidia in other fruit fly species. The observation of compartmentation of CTP synthase now permits a broad range of questions to be addressed concerning not only the structure and function of cytoophidia but also the organization and regulation of CTP synthesis.

Mitochondrial biogenesis and dynamics in the developing and diseased heart.

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The mitochondrion is a complex organelle that serves essential roles in energy transduction, ATP production, and a myriad of cellular signaling events. A finely tuned regulatory network orchestrates the biogenesis, maintenance, and turnover of mitochondria. The high-capacity mitochondrial system in the heart is regulated in a dynamic way to generate and consume enormous amounts of ATP in order to support the constant pumping function in the context of changing energy demands. This review describes the regulatory circuitry and downstream events involved in mitochondrial biogenesis and its coordination with mitochondrial dynamics in developing and diseased hearts.

Impacts of high ATP supply from chloroplasts and mitochondria on the leaf metabolism of *Arabidopsis thaliana*.

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Chloroplasts and mitochondria are the major ATP producing organelles in plant leaves. *Arabidopsis thaliana* purple acid phosphatase 2 (AtPAP2) is a phosphatase dually targeted to the outer membranes of both organelles and it plays a role in the import of selected nuclear-encoded proteins into these two organelles. Overexpression (OE) of AtPAP2 in *A. thaliana* accelerates plant growth and promotes flowering, seed yield, and biomass at maturity. Measurement of ADP/ATP/NADP(+)/NADPH contents in the leaves of 20-day-old OE and wild-type (WT) lines at the end of night and at 1 and 8 h following illumination in a 16/8 h photoperiod revealed that the ATP levels and ATP/NADPH ratios were significantly increased in the OE line at all three time points. The AtPAP2 OE line is therefore a good model to investigate the impact of high energy on the global molecular status of *Arabidopsis*. In this study, transcriptome, proteome, and metabolome profiles of the high ATP transgenic line were examined and compared with those of WT plants. A comparison of OE and WT at the end of the night provide valuable information on the impact of higher ATP output from mitochondria on plant physiology, as mitochondrial respiration is the major source of ATP in the dark in leaves. Similarly, comparison of OE and WT following illumination will provide information on the impact of higher energy output from chloroplasts on plant physiology. OE of AtPAP2 was found to significantly affect the transcript and protein abundances of genes encoded by the two organellar genomes. For example, the protein abundances of many ribosomal proteins encoded by the chloroplast genome were higher in the AtPAP2 OE line under both light and dark conditions, while the protein abundances of multiple components of the photosynthetic complexes were lower. RNA-seq data also showed that the transcription of the mitochondrial genome is greatly affected by the availability of energy. These data reflect that the transcription and translation of organellar genomes are tightly coupled with the energy status. This study thus provides comprehensive information on the impact of high ATP level on plant physiology, from organellar biology to primary and secondary metabolism.

The origin of cytosolic ATP in photosynthetic cells.

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In photosynthetically active cells, both chloroplasts and mitochondria have the capacity to produce ATP via photophosphorylation and oxidative phosphorylation, respectively. Thus, theoretically, both organelles could provide ATP for the cytosol, but the extent, to which they actually do this, and how the process is regulated, both remain unclear. Most of the evidence discussed comes from experiments with rapid fractionation of isolated protoplasts subjected to different treatments in combination with application of specific inhibitors. The results obtained indicate that, under conditions where ATP demand for photosynthetic CO₂ fixation is sufficiently high, the mitochondria supply the bulk of ATP for the cytosol. In contrast, under stress conditions where CO₂ fixation is severely limited, ATP will build up in chloroplasts and it can then be exported to the cytosol, by metabolite shuttle mechanisms. Thus, depending on the conditions, either mitochondria or chloroplasts can supply the bulk of ATP for the cytosol. This supply of ATP is discussed in relation to the idea that mitochondrial functions may be tuned to provide an optimal environment for the chloroplast. By balancing cellular redox states, mitochondria can contribute to an optimal photosynthetic capacity.

The Interface Between ER and Mitochondria: Molecular Compositions and Functions.

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Mitochondria and endoplasmic reticulum (ER) are essential organelles in eukaryotic cells, which play key roles in various biological pathways. Mitochondria are responsible for ATP production, maintenance of Ca^{2+} homeostasis and regulation of apoptosis, while ER is involved in protein folding, lipid metabolism as well as Ca^{2+} homeostasis. These organelles have their own functions, but they also communicate via mitochondrial-associated ER membrane (MAM) to provide another level of regulations in energy production, lipid process, Ca^{2+} buffering, and apoptosis. Hence, defects in MAM alter cell survival and death. Here, we review components forming the molecular junctions of MAM and how MAM regulates cellular functions. Furthermore, we discuss the effects of impaired ER-mitochondrial communication in various neurodegenerative diseases.

Mitochondria in human oogenesis and preimplantation embryogenesis: engines of metabolism, ionic regulation and developmental competence.

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Mitochondria are the most abundant organelles in the mammalian oocyte and early embryo. While their role in ATP production has long been known, only recently has their contribution to oocyte and embryo competence been investigated in the human. This review considers whether such factors as mitochondrial complement size, mitochondrial DNA copy numbers and defects, levels of respiration, and stage-specific spatial distribution, influence the developmental normality and viability of human oocytes and preimplantation-stage embryos. The finding that mitochondrial polarity can differ within and between oocytes and embryos and that these organelles may participate in the regulation of intracellular Ca^{2+} homeostasis are discussed in the context of how focal domains of differential respiration and intracellular-free Ca^{2+} regulation may arise in early development and what functional implications this may have for preimplantation embryogenesis and developmental competence after implantation.
